Quantum-Dot-Based Direct X-Ray Detector for Mammography

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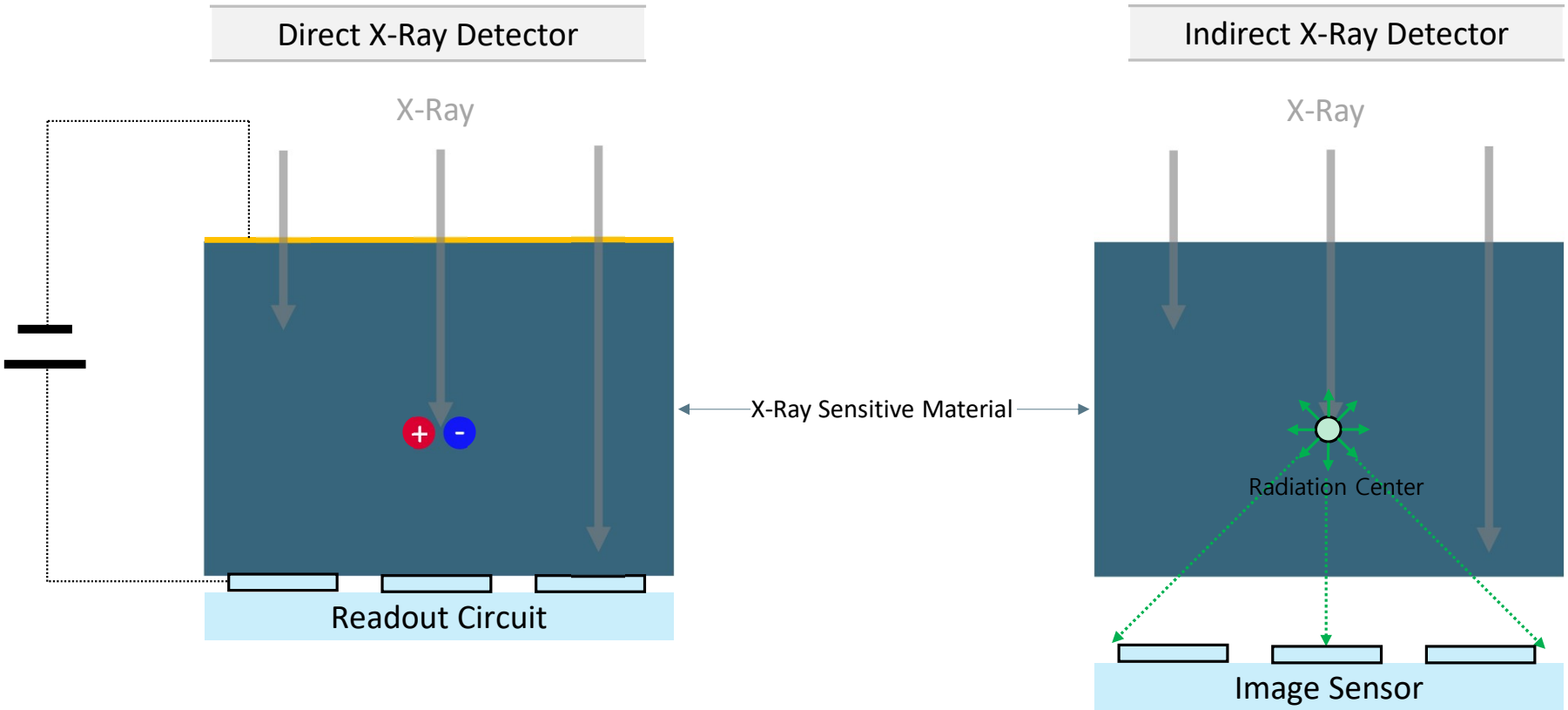
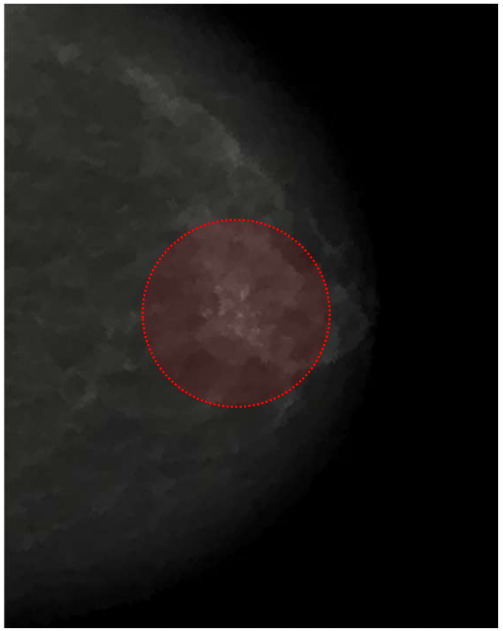
1. Introduction: Classification for X-Ray Detector



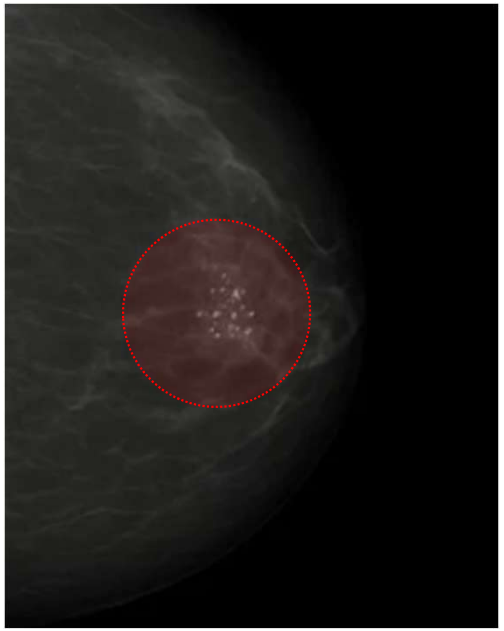
Indirect X-Ray detectors are well-established; but suffer from limited spatial resolution

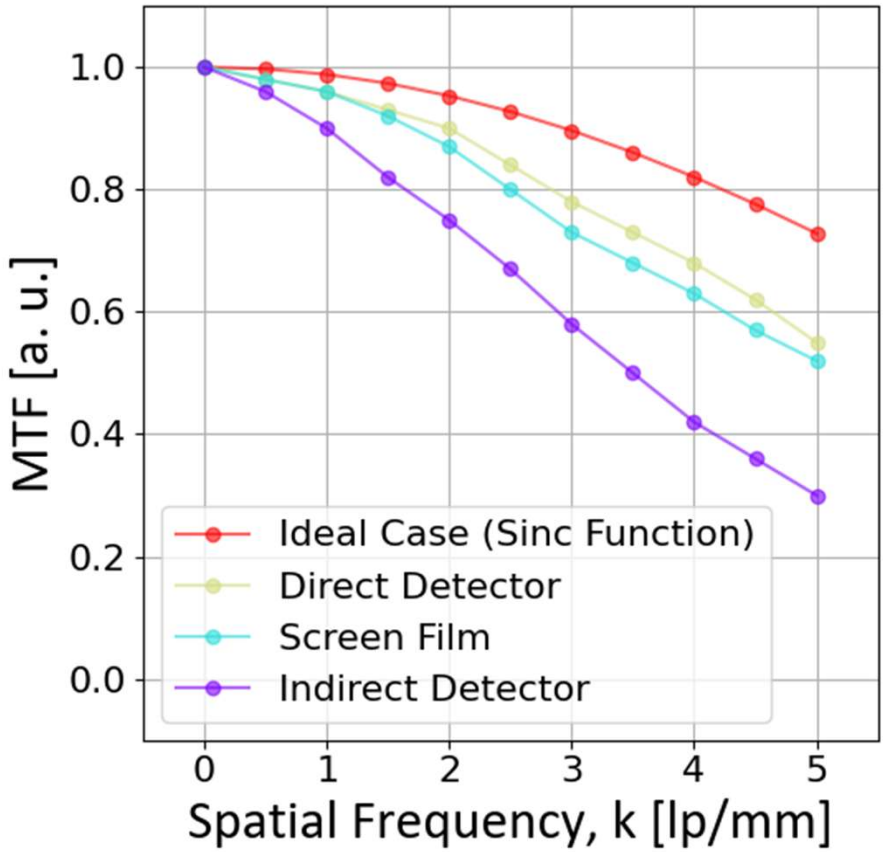
1. Introduction: Requirements for Higher Spatial Resolution

Low Resolution



High Resolution



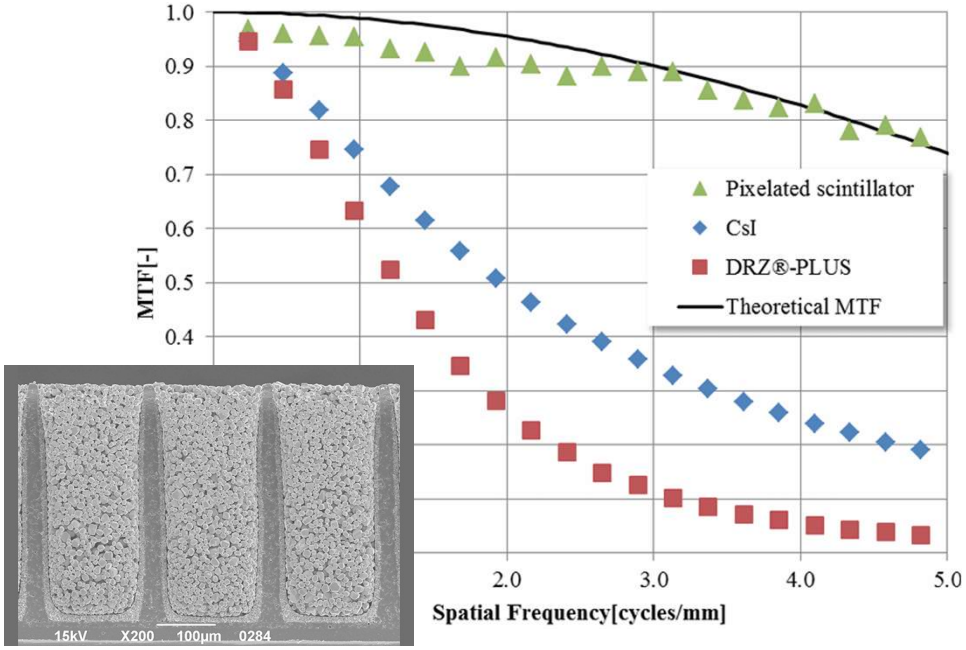


Microcalcification (Sign of Breast Cancer)

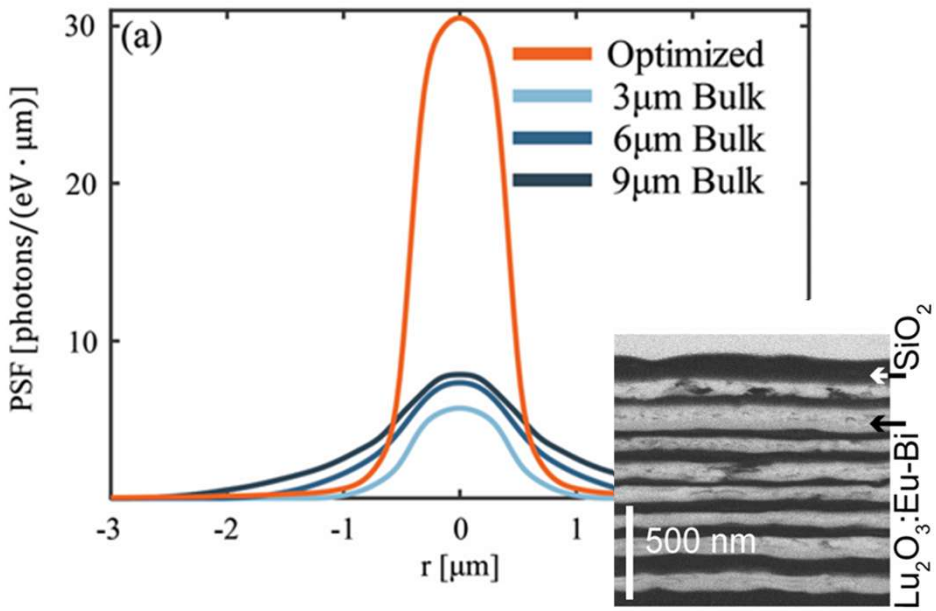
[1] Why High-Brightness Monitors Matter in Mammography
[2] Yaffe, Martin J. "Detectors for digital mammography." *Digital mammography*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2010. 13-31.

1. Introduction: Previous Research (Indirect X-Ray Detectors)

Pixelated Scintillator [1]



Nanophotonic Scintillator [2]



Advanced method allowing for improving spatial resolution; but difficult to commercialize

[1] Miyao, Sho, Takahiro Tanino, and Kazuki Shigeta., *Transactions of The Japan Institute of Electronics Packaging* 11 (2018): E18-011.
[2] Shultzman, Avner, et al., *Advanced Optical Materials* 11.8 (2023): 2202318.

1. Introduction: Previous Research (Direct X-Ray Detectors)

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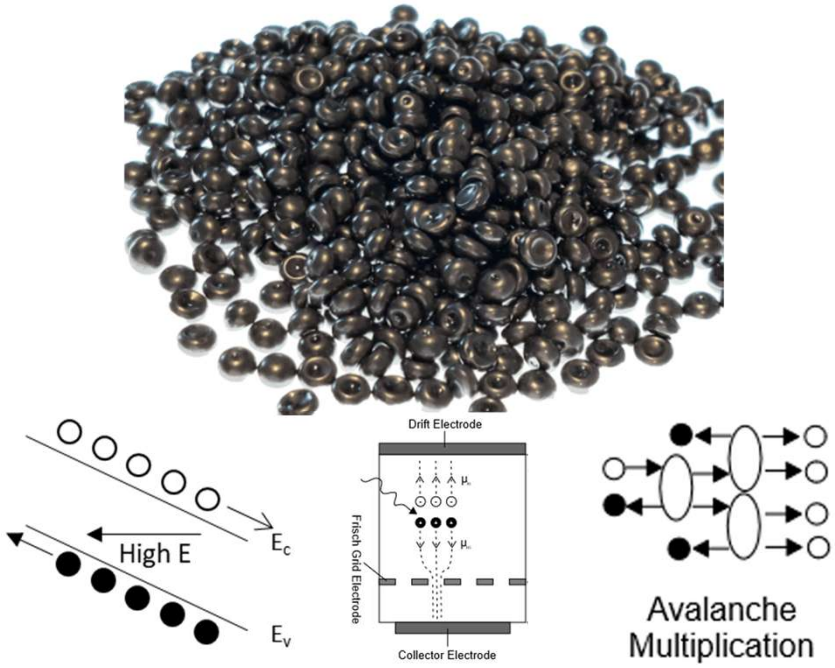
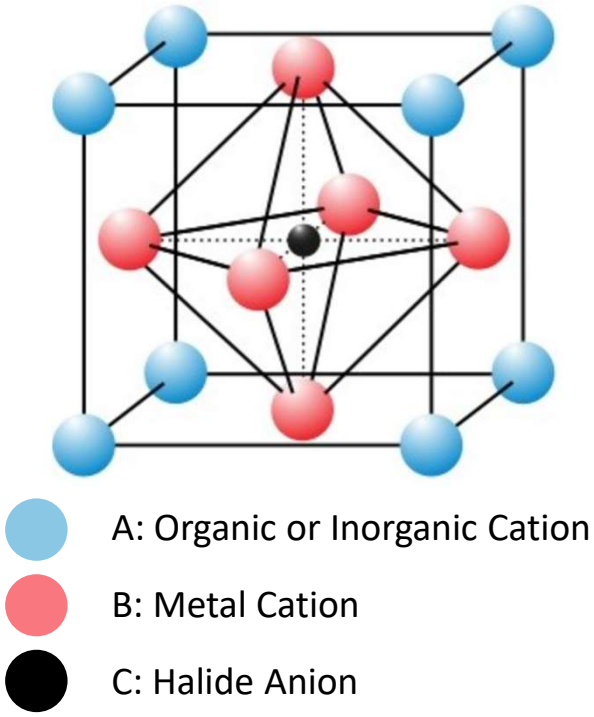
Selenium [1], [2]



Perovskite



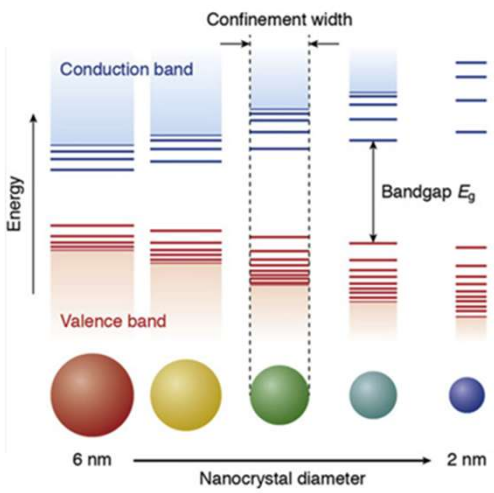
Direct X-Ray conversion allowing for higher spatial resolution and commercialized; but existence of limitations

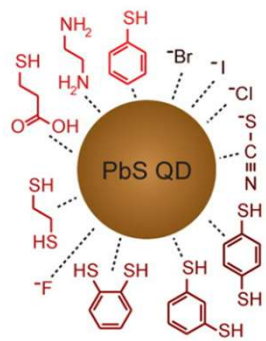
[1] Kasap, Safa, et al., *Physica status solidi (b)* 246.8 (2009): 1794-1805.
[2] Goldan, A. H., et al., *Journal of Applied Physics* 113.22 (2013).

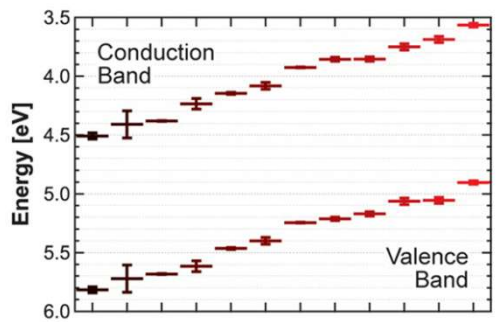
1. Introduction: Quantum Dots

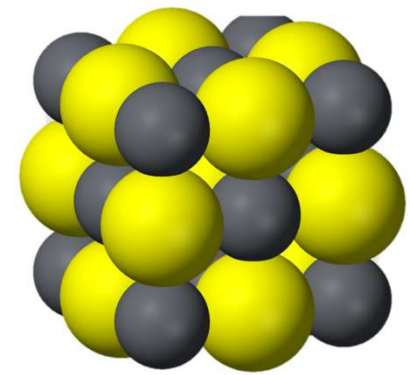
Energy Band Design [1], [2]

High-Z Materials









Bandgap ∝ Size^{-1}

Electron Affinity Control

Pb (Z = 82)

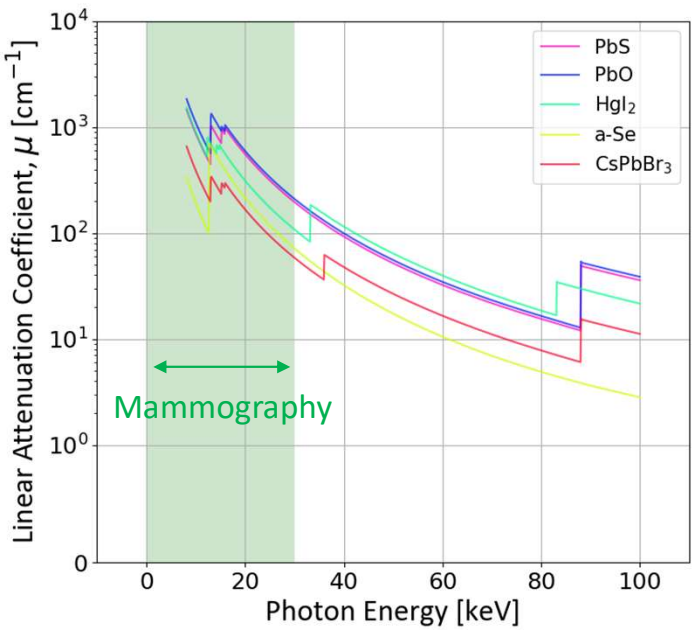
S (Z = 16)

PbS Quantum dots: Matured technology in infrared applications; but rare utilized in X-Ray regime

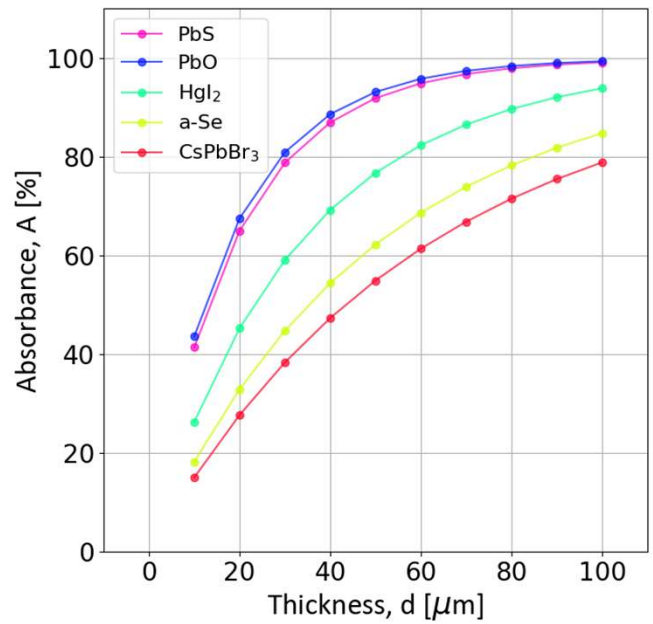
[1] de Mello Donegá, Celso., *Chemical Society Reviews* 40.3 (2011): 1512-1546.
[2] Brown, Patrick R., et al. "Energy level modification in lead sulfide quantum dot thin films through ligand exchange." *ACS nano* 8.6 (2014): 5863-5872.

2. Methodology: Absorption Properties

Attenuation Coefficients [1]



Absorbance vs. Thickness



Lead based materials exhibit superior attenuation coefficient within mammography regime

[1] SpekPy Web

2. Methodology: Sample Information

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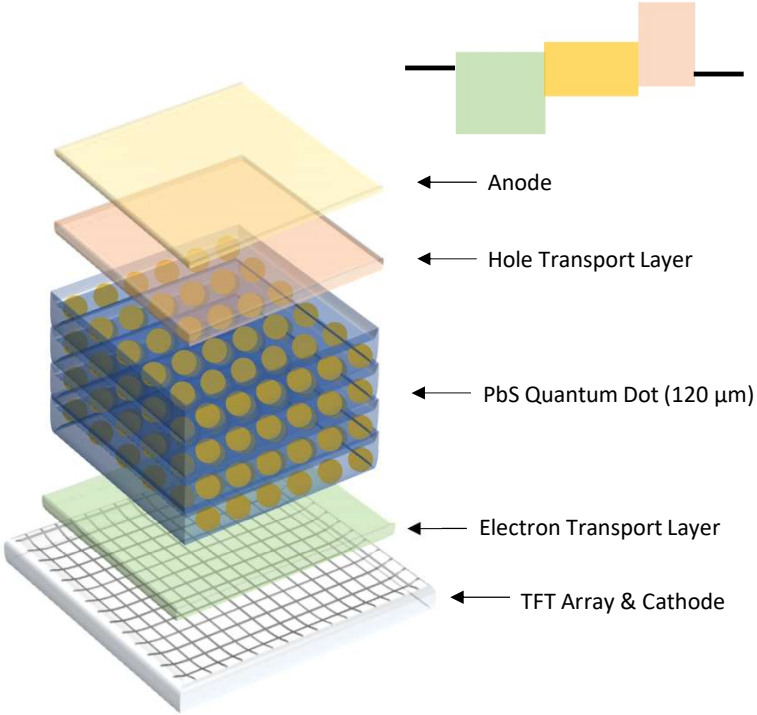
Sample Structure



Sample Image

Optimized photon absorbing layers and transporting layers fabricated on TFT array

2. Methodology: Evaluation Condition

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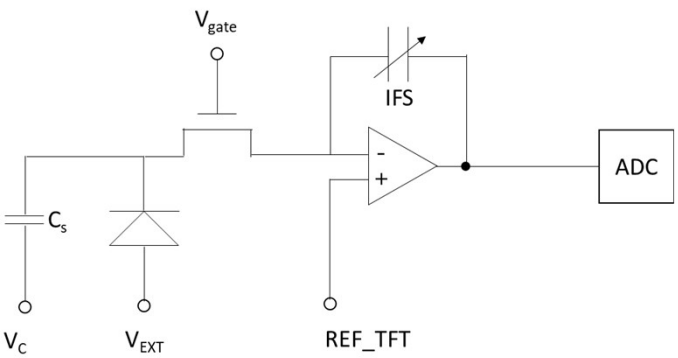
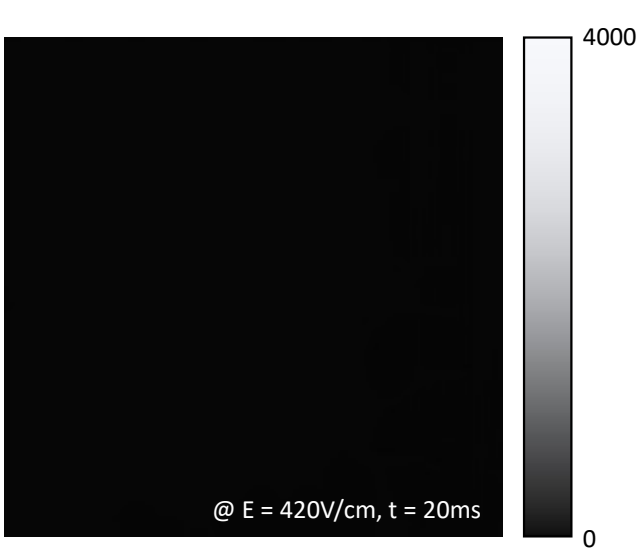
TFT Array	Blank Image	Mammography [1]



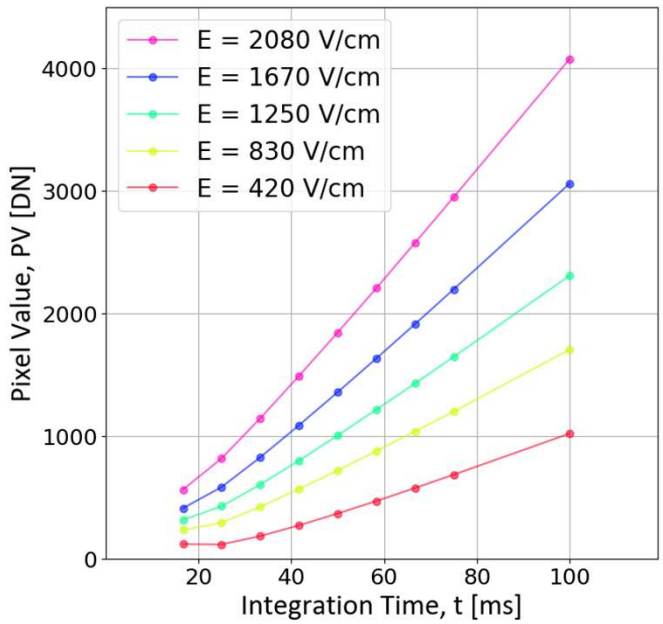


Evaluation without high voltage under standard mammographic evaluation condition

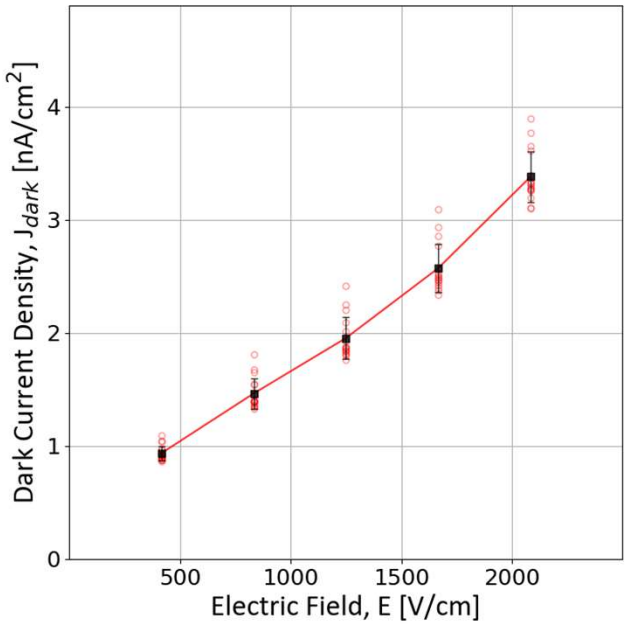
[1] IEC 62220-1-2:2007 | IEC Webstore

3. Results: Dark Current Density

Linearity



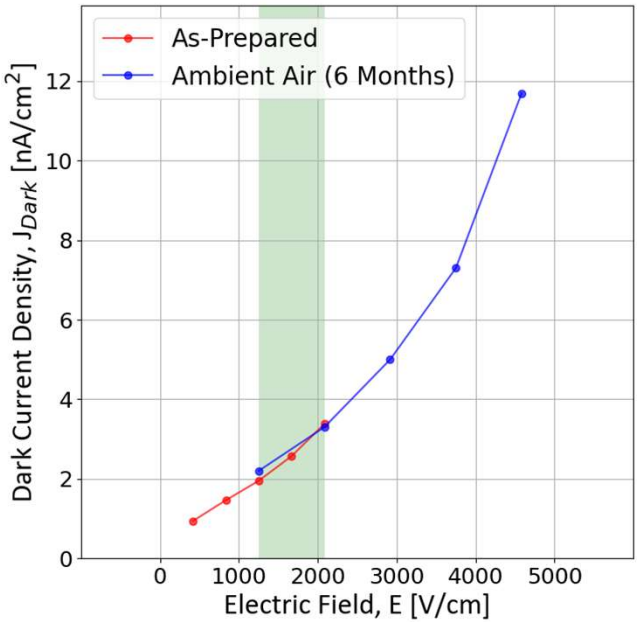
Dark Current Density



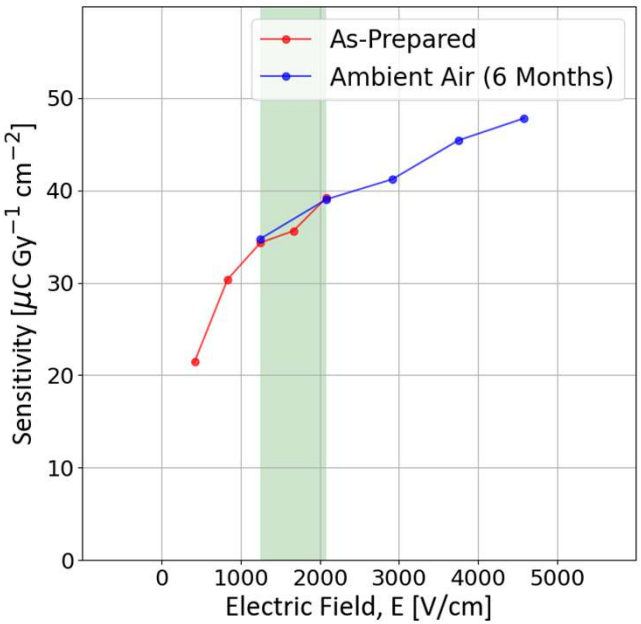
Linearity exists; Dark current density in the order of 1 nA/㎠; Dark signal non-uniformity < 9.3%

3. Results: Environmental Stability

Dark Current Density



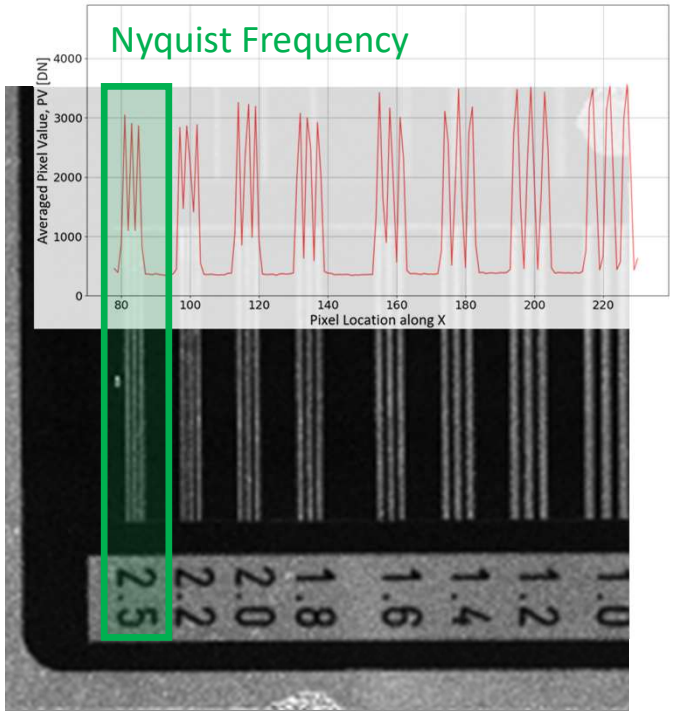
Sensitivity



Long termly stable (over 6 months) in the ambient air both for dark current density and sensitivity

3. Results: Spatial Resolution

Contrast Modulation



Contrast modulation demonstrates sensor spatially resolved up to Nyquist frequency

4. Summary

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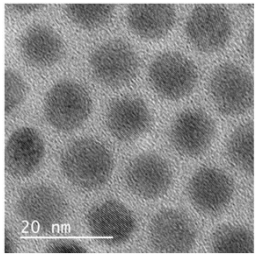
- Achieved quantum dot based mammographic X-Ray detectors
- Available with relatively **thin** & **low voltage operation**
- **Sensitivity higher than commercial a-Se** detectors with dark current density of 1.5nA/㎠
- **Robust to ambient air** condition for over 6 months

	Thickness [μm] @ A= 95%	E_g [eV]	$W_{\pm, Klein}$ [eV]	J_d [nA/cm^2]	Sensitivity [μC/Gycm^2]	E [V/cm]	Stability Issue	References
PbS	**60**	0.9	3.0	1.46	**31**	**8.3×10^2**	**Robust**	This Report
a-PbO	57	1.9	5.8	3.00	N/A	1.0×10^5	Degradation	[1], [2]
HgI_2	106	2.1	6.4	3.00	200	8.0×10^3	Vaporization	[2], [3]
a-Se	166	2.2	6.7	1.00	**20**	1.0×10^5	Crystallization	[2]
$MAPbI_3$	202	2.1	6.3	10^4	2500	2.0×10^3	Degradation	[4]

[1] Grynko, Oleksandr, and Alla Reznik, Photoconductivity and Photoconductive Materials: Fundamentals, Techniques and Applications 2 (2022): 643-688
[2] Kasap, Safa O., and John A. Rowlands., Proceedings of the IEEE 90.4, 591-604 (2002)
[3] Su, Zhong, et al., Physics in Medicine & Biology 50.12 (2005): 2907
[4] Shrestha, Shreetu, et al., Nature Photonics 11.7 (2017): 436-440

4. Further Research



Materials: $\mu\tau$, Ionization Energy

· $\mu\tau$ & ionization energy optimization through dot size, interdot spacing

· Material research to reduce photoluminescence

Devices: Sensitivity, Dark Current, Time Response, Depth Dependency

· Sensitivity optimization through applied bias and thickness

· Dark current suppression through band alignment

· Consideration of time response

· Consideration of depth dependency

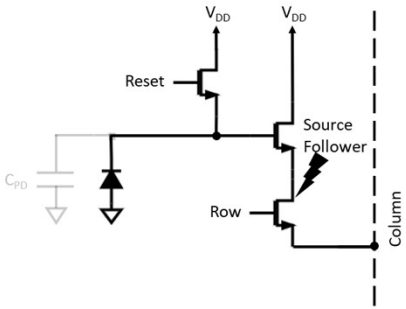


Image Sensors: Pixel Pitch, Temporal Noise

· Reduce pixel pitch for higher resolution mammographic sensors

· Introduce CMOS image sensors to improve image quality

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Thanks